1. Cybersecurity involves the security of
    a. Hardware and software
    b. Software and data
    c. Hardware, software, and data
    d. Hardware, software, data and processes

Answer: c

Justification: Refer to slide 3, lecture 1, week 4

2. _____ security represents the set of policies that monitor and prevent unauthorized access, safeguard the data and network from network traffic. Fill in the blank.
    a. Information
    b. Hardware
    c. Software
    d. Network

Answer: d

Justification: Refer to the book Introduction to Industrial Internet of Things and Industry 4.0, page 66.

3. Locky is an example of which type of cybersecurity threat?
    a. Ransomware
    b. Malware
    c. Phishing
    d. All of the above

Answer: a

Justification: Refer to slides 11-12, lecture 1, week 4

4. What does C represent in the CIA-Triad in cybersecurity?
    a. Cyber-physical
    b. Centralized
    c. Confidentiality
    d. Critical

Answer: c

Justification: Refer to slides 15-16, lecture 1, week 4

5. What is ubiquitous sensing in IIoT?
    a. In IIoT, sensors and actuators are used to control critical elements.
    b. In IIoT, sensors and actuators are used almost everywhere to control, enhance and optimize various functions.
    c. In IIoT, advanced analytics helps to enhance the working condition, increase machine life and optimize operational efficiency.
    d. None of the above

Answer: b

Justification: Refer to slides 5-6, lecture 2, week 4

6. Which of the following is a data integration challenge in deploying IIoT?
    a. Understanding the generated data for analysis and application in business.
    b. Storing the data
    c. Lack of skills
    d. Cybersecurity

Answer: a

Justification: Refer to slides 8-10, lecture 2, week 4

7. Commercialization and the mass production of steam engines marked the beginning of the_____ stage of the Industrial revolution. Fill in the blank.
    a. Fourth
    b. Third
    c. Second
    d. First

Answer: d

Justification: Refer to slide 5, lecture 3, week 4

8. What were the negative effects of Industrial revolution?
    a. Harm to environment by waste products
    b. Inefficient
    c. Bad working environment
    d. All of the above

Answer: d

Justification: Refer to slide 7, lecture 3, week 4

9. How many key elements are there in the Industrial internet?
    a. 4
    b. 3
    c. 5
    d. 2

Answer: b

Justification: Refer to slide 10, lecture 3, week 4

10. How does sensing contributes to the industry?
    a. Increase downtime
    b. Increase cost
    c. Higher degree of automation
    d. Safety hazards

Answer: c

Justification: Refer to slide 3, lecture 4, week 4

11. A smart sensor can perform multiple sensing, unlike the traditional sensors. True or false?
    a. True
    b. False

Answer: a

Justification: Refer to slide 10, lecture 4, week 4

12. What is self-decision making in a smart sensor?
    a. It can self-monitor its operation and changes in the ambience by taking proper decision by itself.
    b. Reduce the cost by reduced downtime
    c. Communicate data
    d. Enable multiple parameter sensing

Answer: a

Justification: Refer to slide 11, lecture 4, week 4

13. What is the full form of SCADA?
    a. Supervisory Confidential and Data Acquisition
    b. Supervisory Control and Data Acquisition
    c. Smart Control and Data Acquisition
    d. Smart Controlled and Direct Acquisition

Answer: b

Justification: Refer to slide 24, lecture 4, week 4

14. What does the Cognitive layer in the 5C architecture for Cyber-physical Systems represents?
    a. System for supervised control: Self-configure, Self-Optimize, Self-Adjust
    b. Sensor records machine health data: data to information interpretation
    c. System for optimized decision: Human-readable interpretation & data visualization
    d. All of the above

Answer: c

Justification: Refer to slide 9, lecture 5, week 4

15. How can smart factory be beneficial in inventory tracking?
    a. Global inter-connectivity of smart factory facilitates real-time updates of inventory
    b. Ease of process re-adjustment facility
    c. End-to-end production line management with sensors
    d. None of the above

Answer: a

Justification: Refer to slide 17, lecture 5, week 4